R^d Visualization

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ABSTRACT

Visualizing high-dimensional data remains a significant challenge due to issues of clutter, interpretability, and scalability. Our approach introduces a novel visualization technique that leverages subspace structures and statistical measures—such as variance and the Gini index—to guide the plotting of data points. While existing techniques such as parallel coordinates, circle segments, and dimensionality reduction methods (e.g., PCA, t-SNE) have provided some insight, they often suffer from information loss, clutter, or limited interpretability and implementation. Our proposed method, Subspace-quadrant based visualization, offers a compact and insightful alternative. It visualizes high-dimensional data using subspace-based quadrant partitioning, color-coded bit vectors, and a circular layout to enhance clarity and support pattern recognition. Subspace-quadrant based visualization supports exploration of both global and local structures, scales effectively up to moderate dimensions (e.g., 10D), and offers an optional linear view to improve interpretability.

Index Terms: Data visualization, high-dimensional data, subspace analysis, variance, Gini index, quadrant visualization.

1 INTRODUCTION

Visualizing high-dimensional datasets is a well-known challenge in data science and analytics. As the number of dimensions increases, traditional visualization techniques often become less effective due to clutter and the difficulty of perceiving complex patterns. This has led to various attempts at reducing dimensionality or enhancing the representation of high-dimensional structures through more intuitive visual metaphors.

Our method aims to address the shortcomings of existing visualizations by introducing a quadrant-based circular plotting technique that captures structural features in subspaces of increasing dimensionality. By combining variance or Gini index-based ordering with bit-vector representations and color-coded regions, our approach provides an interpretable and scalable solution for high-dimensional data visualization. It also prevents information loss and efficiently helps the user interpret the spatial position of the points.

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2 RELATED WORK

Existing approaches to high-dimensional data visualization primarily fall into two categories: dimensionality reduction techniques and direct visualization methods.

Dimensionality reduction techniques such as PCA, t-SNE, and UMAP are commonly used to project high-dimensional data into 2D or 3D space. While effective in preserving either global (PCA) or local (t-SNE, UMAP) structures, these methods can lead to information loss and may obscure important subspace relationships.

Direct visualization techniques include:

- **Parallel Coordinates** [1]: The Parallel Coordinate Evaluation survey provides context on a standard technique. PC represents dimensions as parallel vertical axes and data points as lines connecting values across axes. The survey highlights the importance of user-centered evaluation for variations (axis layouts, clutter reduction) and practical applications.

- **Circle Segments** [2]: A pixel-per-value technique designed for large, high-dimensional datasets. It maps dimensions to circular segments and arranges data values within them, leveraging color mapping for intuition. Strengths include handling more dimensions than screen-width-limited linear techniques and offering interactive dimension rearrangement.

- **Heidi Matrix** [3]: Visualizes high-dimensional index structures (like R-Trees) using a 2D matrix layout. Rows/columns are ordered based on the R-Tree hierarchy. Cell color encodes the "closeness" (based on kNN) of point pairs across various subspaces, revealing index structure and subspace relationships simultaneously.

- **SpaceMAP** [4]: Addresses crowding by explicitly aiming to match the "capacity" (volume) of high- and low-dimensional spaces. It uses the concept of Equivalent Extended Distance (EED) to transform high-dimensional distances before calculating similarities. Key features include hierarchical modeling (near/middle/far fields), estimation of intrinsic dimensionality, and a connection to contrastive learning loss functions (similar to UMAP).

Each of these methods has its own strengths but fails to offer a clean, interpretable, and scalable solution for very high-dimensional data (e.g., 20D, 50D).

3 Strengths of Existing Work

Despite their limitations, current techniques offer various advantages:

- **Parallel Coordinates**[1]: Simple to learn and effective for moderate dimensions; allows direct comparison between dimensions.
- **Circle Segments**[2]: Maps each dimension to a segment and encodes data values using color, suitable for large datasets.
- **Heidi Matrix**:[3] Enables visualization of kNN-based closeness and index structures; captures subspace relationships.
- **PCA and t-SNE**: Widely adopted dimensionality reduction tools—PCA captures global variance, while t-SNE excels at local neighborhood preservation.

4 Limitations of Related Work and Our Contribution

Although existing methods have made progress, they often:

- Fail to preserve both global and local data structures.
- Struggle with interpretability for non-expert users.
- Are not scalable to higher-dimensional datasets.
- Result in cluttered, non-informative plots.

Our contributions address these issues through:

- **Subspace-aware Visualization**: A quadrant-based approach using bit-vector encoding for each subspace, with visual encoding of sign information (positive/negative) through color saturation.
- **Circular Plot Structure**: Each subspace is represented as a concentric ring, with sectors indicating quadrants formed by bit-vector combinations. Sector sizes may be uniform or scaled based on quadrant occupancy.
- **Edge-based Linking**: Data points are plotted in each subspace ring and connected across rings via colored edges indicating class labels, enhancing cluster and transition analysis.
- **Scalability and Interpretability**: Tested on up to 10-dimensional data, with an emphasis on reduced clutter and improved exploratory capabilities.

5 Our Approach

5.1 Sector-Based Circular Visualization

Our core visualization technique follows these steps:

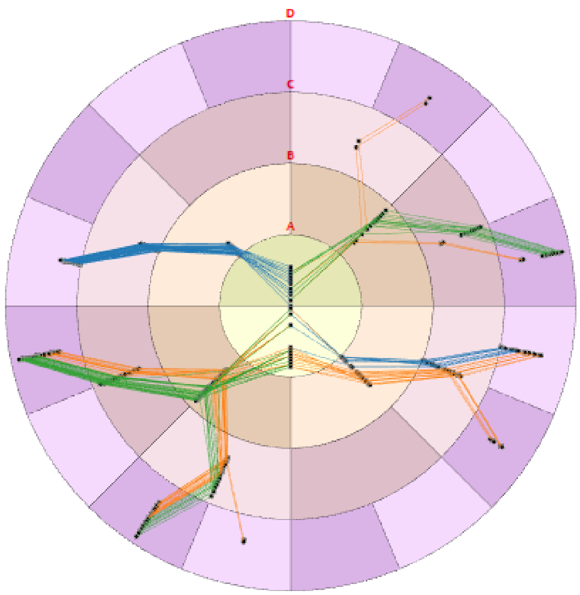
- **Preprocessing**: Normalize the data (scale the data between -1 to 1) and clean the dataset.
- **Subspace Construction**: Create subspaces by sequentially adding dimensions, ordered by decreasing value of variance or Gini index of the features.
- **Bit Vector Conversion**: Convert each data point into binary bit based on the sign of each component (0 for negative, 1 for non-negative).
- **Ring Structure**: Represent each subspace as a concentric ring, moving outward with increasing dimensionality (lower variance or lower gini-index).
- **Quadrant Division**: Each ring is divided into 2^d sectors, where d is the subspace dimension.
- **Color Saturation Encoding**: Sectors are colored based on the signs (bit values) of the quadrant, with light/dark hues indicating non-negative/negative bits.
- **Data Point Plotting**: Data points are placed in the corresponding sectors based on their bit vectors. Each point is plotted once per ring. The coordinate of the data point of the respective subspace is plotted in the respective ring and respective quadrant.
- **Placement of point**: The data point in each sector is placed in the center of the sector to avoid clutter in the visualization and make it easier for the user to understand and interpret the position of the points in the subspaces and their quadrants. To understand the positioning of the points based on their values, a linear visualization is provided along with the subspace-quadrant visualization to avoid clutter and cognitive load on the user.
- **Edge Linking**: All representations of a single data point across subspaces are linked via colored edges. Edge color encodes the class label.

There are two possible views in which we can obtain the visualization, namely:

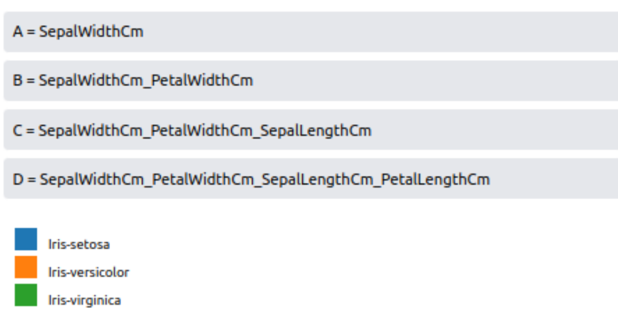
- Uniform/normal View
- Proportional View

where the uniform/normal view and proportional view is based on the area/distribution of the quadrants in each ring.

5.1.1 Uniform quadrants



(a) Uniform Sector view of the Subspace-quadrant based visualization



(b) Legend: Subspace ('_'-separated features) and label colors

Figure 1: Iris Dataset Subspace-quadrant Visualization [4-dimensional]

In the Uniform/Normal view, each sector within a ring is allocated an equal angular area, resulting in a uniform distribution across all rings. This consistent sector sizing helps users easily interpret the spatial distribution of data points and their associated class labels. By standardizing the visual layout, this view clearly highlights clustering patterns and density variations, enabling users to quickly identify which quadrants are highly occupied and potentially more significant or informative. Normal view is the default view when the user runs a dataset to produce visualization.

5.1.2 Non-Uniform quadrants

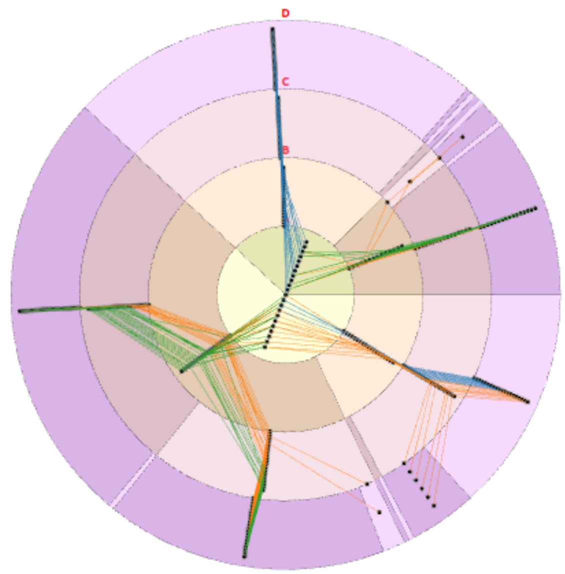
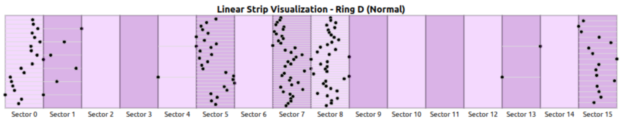


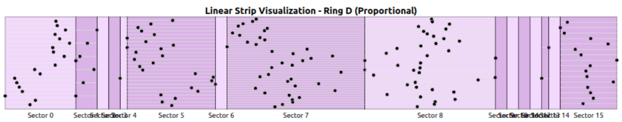
Figure 2: Uniform Sector view of the Subspace-quadrant based visualization

In the non-uniform/proportional view, each sector within a ring is allocated and angle which is proportional to the number of points in that sectors. The The non-uniform sector sizing produces a visualization which makes it easier for the user to quickly identify the most occupied quadrants. For large datasets with d dimension, the last ring will have 2^d sectors. In order to fill all the sectors minimum 2^d points are required which is an exponentially large number for a higher d. Therefore, we observe in high dimensional data that many sectors remain unoccupied. The proportional-sector view allocates a minute angle to the empty sectors which helps identify the unused sectors as well. Since the last ring has 2^d sectors i.e uniformly $2^d/360$ degree for each sector, the proportional view helps the user in this case to eliminate the empty sectors by allocating very small angle and ensuring that the valuable and occupied sectors get more angle, increasing the visibility and insights from the data.

5.2 Linear Visualization for Last Ring



(a) Uniform Sectors in the linear visualization of Ring D



(b) Non-Uniform Sectors in the linear visualization of Ring D

Figure 3: Iris Dataset Linear strip Visualization of external ring [4-dimensional]

To enhance interpretability of the location and plotting of the datapoints, we present a linear visualization of the outermost ring:

- The final ring's sectors are laid out linearly as rectangular strips.
- Each sector is divided horizontally into k arcs (lines) representing individual data points where k is the number of data points in that sector.
- Data points are placed along the arc based on their actual value, with leftmost and rightmost representing the sector's value range.
- In the linear visualization also, we obtain two views, namely, the normal uniform view and the proportional view to make it more intuitive for the user to understand the distribution of the points in the sector along with the most occupied sector.

This view aids in understanding data distribution within sectors and assists in interpreting quadrant occupancy.

6 OUTPUT

The output is an **interactive, layered visualization** designed to help users intuitively explore complex, high-dimensional datasets. It enables the analysis of **relative spatial positioning**, **clustering patterns**, **transitions from one subspace to another** (using rings), and **class distributions** (using colored edges) across multiple subspaces. At its core, the visualization leverages a **360-degree circular layout** that presents a **compact and intuitive** representation of multidimensional data, allowing users to quickly assess how data points are organized and distributed across different hierarchical levels.

To complement the circular view, a **linear view** is integrated, offering a more precise depiction of individual feature distributions and value ranges. This dual-representation approach ensures both **a more intuitive pattern recognition** and **detailed analysis**, making the tool suitable for a wide range of analytical tasks.

Colored edges between corresponding nodes or sectors serve as visual guides, revealing **transitions and relationships** between data points or classes across various subspaces. These connections help in identifying how data evolves or shifts when projected into different dimensions or subspace combinations. The color of the edges represents the label that the point belong to. The cluttering or the movement of the edges helps the user identify the distribution of labels across the dataset.

The key strength of this novel visualization lies in its ability to present complex, high-dimensional information in a **concise and uncluttered manner**. Traditional visualizations often suffer from overplotting and cognitive overload when dealing with high-dimensional data. By organizing data radially and using layered abstraction, this visualization mitigates clutter, maintains visual clarity, and enhances interpretability. Ultimately, it provides users with a clearer, more precise understanding of the underlying dataset structure and behavior.

7 LIMITATIONS

While our method improves clarity and insight for high-dimensional data, some limitations remain:

- May be less effective for very sparse or weakly structured datasets.
- Class overlap and a large number of labels can reduce edge clarity and might interfere in obtaining valuable inferences.
- High-dimensional data increases the number of sectors, which may cause visual clutter. Since the last ring has 2^d sectors i.e uniformly $2^d/360$ degree for each sector, this could be extremely small for higher dimension (d).
- Computational and memory costs increase with dimensionality as the memory used for storing the sectors of each ring along with its data points increase exponentially with increase in dimension.

8 FUTURE WORK

To further develop and refine our method, future directions include:

- **Polar Coordinate Version**: Implement a polar visualization for comparison and to solve the problem of sector distribution in last rings in higher dimensional dataset.
- **Optimization Algorithms**: Improve rendering and computation speed to handle higher dimensions and larger datasets.
- **Scalability Enhancements**: Extend support to higher-dimensional datasets (e.g., ¿20D).

9 CONCLUSION

Visualizing high-dimensional data remains a persistent challenge in data science due to the complexity of capturing structure, interpretability, and interrelationships across multiple dimensions. In this paper, we introduced a novel subspace-quadrant based circular visualization technique that addresses these challenges by organizing high-dimensional data into an intuitive, interpretable, and scalable visual format.

Our method capitalizes on subspace construction guided by statistical metrics such as variance and the Gini index, creating a hierarchy of concentric rings, each representing a subspace of increasing dimensionality. By encoding data point signs as binary bit vectors and mapping them to sectors within each ring, we enable precise visual separation and clustering. The use of color saturation and edge-based linking of points across rings further enhances interpretability and supports class-level and structural pattern discovery.

The visualization supports both a uniform and a proportional sector view, allowing analysts to switch between balanced quadrant representations and occupancy-weighted sector emphasis. Additionally, the linear visualization of the outermost ring offers an alternative way to inspect value distributions and sector-level detail, reducing cognitive load and clutter.

Through this dual-layered approach—circular and linear—we bridge the gap between global data structure awareness and local data point analysis. Our method is interactive, extensible up to moderate dimensions (e.g., 10D), and provides insights that are often lost in traditional dimensionality reduction or overplotted coordinate-based methods.

This work not only introduces a new visualization paradigm but also opens up pathways for more interpretable machine learning, better exploratory data analysis, and interactive data storytelling in high-dimensional contexts. With further enhancements in scalability and computation, this method has the potential to become a foundational tool in the visual analytics toolbox for complex datasets.

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